

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Department of Electrical Engineering

End Semester Examination

Sub: Electronic Devices And Circuits

Semester: IV

Date: 16/05/2015

Time: 10:00 am – 01:00pm

Max. Marks: 70

INSTRUCTIONS TO THE STUDENTS

1. Question No. 1 is compulsory
2. Attempt any FIVE questions from Q2 to Q7
3. Necessary data is given in respective questions. If such data is not given it means that the knowledge of that data is a part of examination.
4. If some part or parameter is noticed to be missing, you may appropriately assume and mention the same clearly.

Que 1. Choose the correct option from the following:

(10)

- i) A transistor is a operated device.
A) Current C) both
B) Voltage D) none of these
- ii) Digital circuit can be made by repeated use of
A) OR C) NAND
B) NOT D) EX-OR
- iii) In Boolean algebra, the plus (+) sign indicates operation.
A) AND C) NOT
B) OR D) none of these
- iv) A zener diode is device.
A) Non-linear C) amplifying
B) Linear D) none of these
- v) In zero signal conditions, a transistor sees load.
A) d.c. C) both a.c. and d.c.
B) a.c D) none of these
- vi) A varactor is usually
A) Forward biased C) unbiased
B) Reverse biased D) none of these
- vii) In general, transistor should have input impedance
A) high C) very low
B) low D) very high
- viii) A differential amplifier
A) Is a part of an OP-AMP C) has 2 outputs
B) Has 1 input and 1 output D) answers A and C
- ix) The output of a particular OP-AMP increases 8V in 12 μ s. The slew rate is
A) 90 V/ μ s C) 1.5 V/ μ s
B) 0.67 V/ μ s D) none of these

- x) A p-n junction diode is device.
- | | |
|--------------------------|-------------------------|
| A) Controlled device | C) bidirectional device |
| B) Unidirectional device | D) none of these |

- Que. 2. A) Explain the process by which majority and minority carriers are produced also briefly describe difference between electron flow and hole flow. (06)
- B) With a neat labelled circuit diagram explain construction and operation of full wave rectifier having a) bridge configuration b) centre tap transformer configuration. (06)
- Que. 3. A) Describe various methods of transistor biasing. (06)
- B) Define in brief transistor biasing. (06)
- Que. 4. A) Give classification of amplifier in brief. (06)
- B) Explain CB connection with characteristics and derive equation for collector current. (06)
- Que 5. A) Explain AC analysis of OP-AMP. (06)
- B) An amplifier has an open-circuit voltage gain of 70 db and an output resistance of 1.5 k Ω . Determine the minimum value of load resistance so that voltage gain is not more than 67db. (06)
- Que 6. A) Give advantages and disadvantages of digital electronics. (06)
- B) Describe NAND gate as universal gate. (06)
- Que 7. A) Explain i) half adder ii) full adder (06)
- B) Write short note on JK flip-flop. (06)

X
